Electromagnetic induction

antocciphysics | TEACHER KEY | CPA

Learning target: Explain why changing magnetic flux can induce an emf.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 7, Day 11; 3 course days later.*
E=Pt=15 J.

02. Explain a worked example (worked)
Motion changes flux; stationary unchanging flux gives zero induced emf.

03. Look and explain (model)
Moving magnet can change flux; the stationary arrangement does not produce sustained emf from motion.

04. Build the solution (practice)
Opposite sign, for unchanged coil and meter connections.

05. Use the idea (reason)
If external flux decreases, the induced field can reinforce its original direction to oppose the decrease.

06. Choose a method (mixed) *Review: Unit 7, Day 2; 12 course days later.*
Force magnitude becomes 4 times as large, from Coulomb’s law.

07. Exit check (exit)
Move faster, increase turns, increase field change or effective area; accept with relevant controls.

08. STRETCH: Apply it another way (stretch)
Greater rate of flux change or more identical turns can increase voltage; hold geometry/field/path or speed fixed for fair comparison.

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01. Bring back an earlier idea (retrieve) *Review: Unit 7, Day 11; 3 course days later.*
V=15 V; P=IV=I²R=75 W.

02. Explain a worked example (worked)
Motion changes flux; stationary unchanging flux gives zero induced emf.

03. Represent and interpret (model)
Moving magnet can change flux; the stationary arrangement does not produce sustained emf from motion.

04. Solve and justify (practice)
.80 V.

05. Analyze the reasoning (reason)
If external flux decreases, the induced field can reinforce its original direction to oppose the decrease.

06. Choose a method (mixed) *Review: Unit 7, Day 2; 12 course days later.*
$F_f/F_i=16/36\approx0.4444$.

07. Exit check (exit)
Four times; same number of turns and same flux change.

08. STRETCH: Apply it another way (stretch)
7.5 V; induced effect opposes the change in flux, not necessarily the existing field itself.